

1 SYSTEM OVERVIEW

Communication ambiguity is recipient-dependent: a message that is clear to one recipient may be unclear to another because their conversational context differs. AOT-GOAT reads the sender–recipient history, detects **recipient-conditioned ambiguity** in a draft message, and makes the smallest edit that removes it **only when needed**. It is meant to help senders writing to people who lack shared context (e.g., a new contact in another time zone). The current implementation is a deterministic rule/lexicon/heuristic pipeline with optional LLM assistance and validation; it is not a trained neural model.

2 INPUT TO OUTPUT



Safeguards: sender intent is preserved (unedited text kept verbatim; negations and numbers unchanged); no unsupported facts are invented (edits use only sender-supplied context, history, or computed dates); an unresolvable span is left as written and the sender is asked.

3 RISK TYPES (IMPLEMENTATION CATEGORIES, NOT DEMOGRAPHIC GROUPS)

Coreference Unclear reference to an entity	Temporal Ambiguous time expression	Scope Unclear range or recipients	Terminology Unclear jargon or acronym	Implicit Context Meaning depends on unstated shared context
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4 DATA & EVALUATION

SYNTHETIC EVALUATION SET – not a real-world dataset

- 59 project-created cases
- 41 development cases (used while building the rules)
- 18 holdout cases (written afterwards, not used for tuning; same author and style)
- One binary label per (case, risk type): does *this* recipient need an intervention? Offline deterministic analyzer.

5 PERFORMANCE

Metric	Dev (n=41)	Holdout (n=18)	Overall (n=59)
Precision	0.929	1.000	0.951
Recall	1.000	0.929	0.975
F1	0.963	0.963	0.963

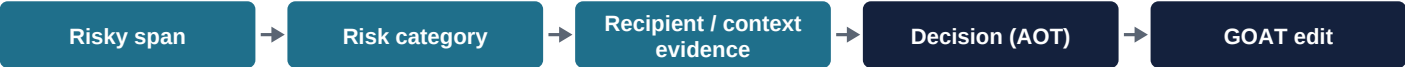
Risk-type level, micro-averaged over (case, risk type) labels.

Decision-level intervention F1

PASS vs INTERVENE | P 0.969 R 0.969
TP 31 TN 26 FP 1 FN 1 | FPR 3.7% FNR 3.1%
Predicted intervention rate: 32/59 = 54.2%

0.969

6 EXPLAINABILITY



AOT-GOAT provides **intrinsic, pipeline-level** explanations: every intervention is traceable through the chain above (the output reports each span's type, reason, evidence, risk score, decision and edit) rather than a bare, unexplained prediction. No post-hoc neural explainability (SHAP, LIME, attention) is used or claimed.

7 BIAS & FAIRNESS CONSIDERATIONS

- The audit examined the risk categories and recipient/context conditions represented by the implementation.
- Demographic fairness analysis was **not performed**: demographic/sensitive attributes are not present in the evaluation basis.
- Per-risk and per-context quantitative subgroup metrics could **not be reconstructed**: row-level prediction/context outputs were not retained.
- No claim is made that the system is unbiased or fair.

8 LIMITATIONS

- Small synthetic evaluation set, written by the same author as the rules
- No retained row-level outputs for detailed subgroup analysis
- English-only rule/lexicon/heuristic prototype; misses paraphrase and unmarked implicit context; some idiom false positives
- Results may not generalize to real-world communication
- Prototype, not a production-scale deployment

INTENDED USE | A research/educational prototype for recipient-aware ambiguity detection and minimal message adaptation. It should not be presented as a production communication-safety system or as a validated real-world fairness system.